

Reproducibility as a Structural Basis for Cognition:

From Physical Laws to Representational Algorithms

Abstract

This paper proposes a structural foundation for reinterpreting the core concepts of cognition, information, and language—such as known/unknown states, algorithms, and symbolic operations—through the lens of **reproducibility**.

We begin by reframing reproducibility in the physical domain as the **stability of state transitions** driven by the interplay of **structure, interaction, and energy**. Structure constrains the space of possible interactions; interactions restrict the sequence of transitions; and energy drives those transitions. Through this triadic relation, physical systems exhibit reproducibility: under identical conditions, they follow identical trajectories.

However, a cognitive agent cannot access physical trajectories directly. It can only operate within the **representational domain**, obtained through biological sensory mappings and internal world mappings. What is projected into the representational domain is not the trajectory itself but only **states**, stripped of their causal transition structure. Consequently, reproducibility in the representational domain emerges as **partial reproducibility**, a juxtaposition of fragments of the reproducibility inherent in the physical domain.

Within this partially reproducible domain, *kichi* (known) and *michi* (unknown) are defined as **entities**, i.e., representational states. A known entity is one that can be reconstructed without breakdown within the representational domain. An unknown entity is one that cannot be reconstructed, or one that includes unobserved states and unconfirmed transition paths permitted by the underlying structure—a hierarchically broader class of entities. Algorithms are redefined as **transition sequences**, i.e., ordered series of states within the representational domain, and symbolic or linguistic operations are understood as reconstructions of such state sequences.

The reproducibility of symbolic and linguistic operations is constrained not by the physical domain but by the **social conventions** of the symbolic system. Mathematical notation is governed by an **axiomatic consensus** that tightly restricts permissible reconstructions, yielding high reproducibility. Natural language, by contrast, is weakly constrained, allowing a far larger space of possible transition sequences and therefore exhibiting substantial variability.

Finally, we show that symbolic and linguistic operations do not float freely as abstract representational processes. They are phenomena grounded in the physical domain, enabled by **energy consumption (ATP)** and **structural constraints (neural circuitry)**. Moreover, partial reproducibility in the representational domain is not a defect but a **structural advantage**, allowing cognitive agents to bypass computational costs and generate new transition sequences—new algorithms.

This paper provides a unified theoretical framework for reconstructing the foundations of cognition, language, and computation through reproducibility, offering a structural basis for treating known/unknown entities, algorithms, and symbolic operations within a single coherent system.

1. Introduction: Reproducibility as the Question Underlying the Implementation of Known/Unknown

The concepts of *known* and *unknown* are widely used across cognitive science, information science, and the life sciences. Yet what “known” actually means remains ambiguous, and no implementable formalization has been established. For example, if a symbol is considered “known,” then symbols carved into books or stone tablets should also count as known; but unless they are reutilized by a subject, they remain mere traces. A word from an unfamiliar foreign language written on paper is not known unless the subject maintains it as a state that can be reutilized. A calculator can manipulate the symbols “+” and “−,” but this does not mean the calculator interprets or reutilizes them as a subject; it merely reacts through fixed circuitry. The existence of a symbol and the subject’s ability to hold that symbol as a reproducible state are entirely different phenomena.

In the Michi-Drive Hypothesis (DOI: 10.5281/zenodo.20990020), *known* was defined as “a state that possesses a reutilizable algorithm.” This definition immediately raises several questions:

- What is reutilization?
- What constitutes an algorithm?
- Under what conditions can a procedure be considered reutilizable?
- Why do some things admit reproducibility while others do not?

To treat known/unknown not as mere labels but as implementable concepts, these questions must be addressed. However, existing frameworks focus on representational aspects—such as the presence of symbols, symbolic manipulation, or the existence of memory—and make it difficult to treat reutilizability and reproducibility as unified physical state transformations.

This paper aims to provide a foundation for implementing known/unknown, algorithms, and reutilizability by reframing the question “What is reproducibility?” in terms of **state transformations driven by structure, interaction, and energy**.

The subject considered in this paper is the cognitive structure of humans; it does not assume autonomous reasoning or external-world action by artificial intelligence.

1.1 Reproducibility as a Perspective: Starting from the Physical Law Domain

Why begin from the physical law domain?

The mere existence of a symbol and the subject’s ability to maintain that symbol as a reproducible state are entirely different phenomena. Addressing this asymmetry requires returning to the most fundamental physical conditions that make reproducibility possible in the first place. To treat known/unknown and algorithmic reutilizability in an implementable manner, it is essential to adopt the physical basis of reproducibility as the starting point.

Anchored in this perspective, this paper first defines reproducibility within the physical law domain. Here, reproducibility refers to the capacity of a physical system’s state transitions to recur under identical conditions. That is, when energy is applied to a physical system with a given structure, interactions drive changes in state, and reproducibility is grounded in the fact that these transitions follow **the same trajectory**.

The term *structure* refers to the configuration of interactions available to the physical system. Structure constrains the space of possible interactions and determines how energy can drive state transitions within that space.

The perspective on reproducibility adopted in this paper does not claim to describe the ultimate truth of the world. Rather, it serves as a **conceptual cut**—a starting point for treating known/unknown, algorithms, and reutilizability in an implementable form. A reductionist bottom-up approach leads to combinatorial explosion in the space of possible state transitions, making it difficult to unify known/unknown and reutilizability. For this reason, the paper begins by grounding reproducibility in the structural constraints of physical systems: how structure bounds interactions, how interactions permit certain transition sequences, and how energy drives those transitions.

The question of how a subject structures the external world into an internal world, and how internal states can be reactivated within the representational domain, was addressed in a separate paper (DOI: 10.5281/zenodo.22870778). In the present paper, reproducibility in the physical domain is adopted as the anchor upon which the partial reproducibility of the representational domain can be reconstructed.

1.2 The Physical Basis of Reproducibility: The Triadic Relation of Structure × Interaction × Energy (Central Thesis of This Paper)

As outlined in Section 1.1, this paper treats reproducibility as state transitions within the physical law domain. To clarify the conditions under which such reproducibility arises, the following central thesis is adopted:

Reproducibility emerges from the triadic relation of

- (1) the finiteness of the possibility space imposed by structure,**
- (2) the constraints on transition sequences imposed by interaction, and**
- (3) the driving force supplied by energy.**

These three elements are indispensable for ensuring that state transitions “follow the same trajectory under the same conditions.” Structure limits the space of possible interactions; interactions determine the causal ordering of transitions; and energy drives those transitions. When these three elements are present, an infinite possibility space is projected into a finite transition space, and state transitions recur as stable trajectories.

- **Structure** → Finite-izes the possibility space of interactions available to a physical system.
- **Interaction** → Provides the causal organization that determines the direction and sequence of state transitions; interactions correspond to concrete physical processes labeled by types such as thermal, vibrational, mechanical, or electromagnetic.
- **Energy** → Functions not as an abstract driving force but as the “cost of transformation” inherent in state change; energy appears inseparably bound to the type of interaction (e.g., heat, vibration).
- **Transition** → The physical change produced through interaction × energy.
- **Reproducibility** → The recurrence of the same transition sequence under identical conditions, constrained by structure and interaction.

This paper positions the **triadic relation of structure × interaction × energy** as the common physical foundation underlying phenomena such as computation, cognition, language, and the distinction between known and unknown.

1.3 Connecting to Known/Unknown: Known/Unknown as Interpretations within the Representational Domain

As outlined in Sections 1.1–1.2, this paper adopts reproducibility in the physical law domain as its starting point. Reproducibility in this domain concerns whether a physical system follows the same trajectory under

identical conditions, governed by the triadic relation of structure, interaction, and energy.

However, known/unknown and algorithmic reutilizability are not concepts that arise in the external world itself. They are concepts that arise within the **representational domain** formed inside the subject. A subject perceives the external world only through internal physical states generated by biological sensory mappings. The physical states of the external world are mapped into internal physical changes via sensory processes, and these internal physical states are further organized into the structural configuration of the representational domain.

Within the representational domain, *known* and *unknown* are determined by whether a representational state can be **reactivated (refrain)**—whether the representation can internally arise again, such as in the replay of a sound or the recall of a visual image.

A crucial point here is that **reproducibility in the physical domain and reproducibility in the representational domain belong to different hierarchical levels:**

- **Non-reproducibility in the physical domain**
→ A physical failure of reproducibility due to undetermined structure, unstable interactions, insufficient energy, etc.
- **Non-reactivation in the representational domain**
→ A representational “unknown,” arising when internal reactivation fails within the subject’s internal world.

These two forms of non-reproducibility must be strictly distinguished.

The aim of this paper is not to foreground known/unknown themselves, but rather to **present the conditions under which reproducibility is established in the physical domain, and to provide a foundation for reinterpreting known/unknown within the representational domain.**

The structuring of the representational domain, the sensory mappings that generate internal states, and the mechanisms of internal reactivation were discussed in detail in DOI: 10.5281/zenodo.22870778. The definition of kichi/michi and the resulting Michi-Drive hypothesis were elaborated in DOI: 10.5281/zenodo.20990020. This paper assumes those results and refers to them only where necessary, leaving their details to those works.

On this basis, the present paper examines how reproducibility in the physical domain—when taken as the foundational layer—leads to reproducibility in the representational domain, i.e., **partial reproducibility**, and how this partial reproducibility enables the reinterpretation of known/unknown, reutilizability, and algorithms.

1.4 Structure of This Paper (Roadmap)

To redefine known/unknown, algorithms, and reutilizability in an implementable form, this paper builds a two-layer structure consisting of the physical law domain and the representational domain. The overall organization is as follows:

- **Chapter 2: The Physical Basis of Reproducibility — Structure × Interaction × Energy**
This chapter clarifies the conditions under which reproducibility is established in the physical law domain and defines the triadic relation (structure, interaction, energy) that serves as the foundation of this paper.

- **Chapter 3: Partial Reproducibility in the Representational Domain — Structural Constraints as Limits of Mapping**

This chapter shows that the world available to the subject is restricted to the representational domain formed through biological sensory mappings and internal world mappings, and it explicates the resulting “partial reproducibility.”

- **Chapter 4: A Structural Definition of Known (kichi) / Unknown (michi)**

Based on the partial reproducibility of the representational domain, this chapter defines known/unknown as *entities* in terms of reconstructibility (refrain).

- **Chapter 5: Connecting to Symbolic and Linguistic Operations**

This chapter reinterprets symbolic and linguistic operations as algorithms, understood as reconstructions of state sequences within the representational domain.

- **Chapter 6: Conclusion — The Theoretical Ground Provided by the Perspective of Reproducibility**

This chapter summarizes how the two-layer structure of the physical and representational domains provides a unified theoretical foundation for explaining known/unknown, algorithms, and symbolic operations through the perspective of reproducibility.

2. The Physical Basis of Reproducibility: Structure × Interaction × Energy

To treat reproducibility as a phenomenon within the physical law domain, it is necessary to understand how state transitions arise and how they stabilize. This chapter presents the physical basis on which reproducibility is established, not as a formal proof but as an explanation of its necessity.

The triadic relation of structure, interaction, and energy discussed in this paper does not subordinate itself to existing dynamical systems theory. Rather, it constitutes an original framework designed to unify computation, cognition, language, and the distinction between known and unknown. However, to aid the reader’s understanding, this chapter refers—only as a minimal translation—to a concept analogous to *phase space* in physics as a guiding auxiliary notion.

2.1 Why Structure Finite-izes the Possibility Space

To treat reproducibility as a phenomenon within the physical law domain, we must first understand the factors that determine **how a physical system can behave**. At the center of this is **structure**.

Although physical systems generally possess many degrees of freedom, these degrees of freedom do not behave without constraint. Their behavior is strongly limited by the **configuration of the possibility space of interactions**—that is, by structure.

Structure can be understood as:

the configuration that determines which components may interact with which other components, and in what ways.

Once this configuration is fixed, the possibility space of state transitions that the system can take is naturally finite-ized. Without structure, state transitions would diffuse into an unbounded space of possibilities, and reproducibility could not arise.

This intuition can be seen in the following examples:

- **Molecular structure** → constrains how atoms can bond with one another.
- **Neural network structure** → constrains how neurons can form chains of activation.
- **Mechanical structure** → constrains how components can transmit motion.

These examples are not meant to subordinate the theory of this paper to existing hierarchical structures. Rather, they serve as intuitive illustrations of how structure functions as a possibility space for interactions.

(Supplement)

The “finite-ization of the possibility space” discussed in this paper can be understood as a phenomenon analogous to constraints on phase space in physics. However, the notion of structure used here does not depend on the mathematical definition of phase space. Instead, it is reinterpreted as a foundational concept applicable to broader systems such as computation, cognition, and language.

2.2 Why Interaction Constrains Transition Sequences

Even if structure finite-izes the possibility space, how a system actually changes is determined by **interaction**. Interaction refers to the physical processes through which components within a structure **causally influence one another**, accompanied by concrete labels such as thermal, vibrational, mechanical, or electromagnetic. Interactions occur only within the possibility space defined by structure, and they instantiate that structure by producing changes in state variables.

Interaction provides the causal organization that determines the **direction** and **sequence** of state transitions:

- Because interactions exist, the direction in which a state changes is determined.
- Because interactions exist, changes appear as **sequences**.
- Because sequences exist, **trajectories** emerge.

Without interaction, energy would merely dissipate, and no directed change would occur. In this sense, interaction is the concretization of structure: it constrains the sequences of state transitions within the bounds permitted by structure.

Without this “constraint of sequences,” state transitions would diffuse in different directions each time, and reproducibility could not arise.

(Supplement)

When structures interact with one another, each structure is treated as a “component” at a higher hierarchical level. Structure is a hierarchy-dependent concept, but this hierarchy does not expand without bound.

2.3 Why Energy Is Used Only Through Interaction

Energy is the driving source that enables changes in state. However, energy alone does not determine the direction of a state transition. Energy merely provides the **possibility of change**, while **interaction** determines *how* the change unfolds.

Interaction refers to the physical processes through which components causally influence one another, accompanied by concrete labels such as thermal, vibrational, mechanical, or electromagnetic. Energy contributes to state change only in forms inseparable from these interaction labels. In other words, energy is

not an abstract driving force; it functions as the **cost of transformation**, the necessary expenditure associated with state change.

- Energy provides only the *possibility* of change.
- Interaction determines *how* the change occurs.
- Without interaction, energy merely dissipates and no trajectory emerges.

Thus, energy becomes a “directed change” only through interaction.

Structure limits the possibility space of interactions,
interaction determines the direction and sequence of transitions,
and energy drives those transitions.

Without this triadic relation, state transitions would diffuse in different directions each time, and reproducibility could not arise.

2.4 Why the Triadic Relation Generates Reproducibility

Reproducibility is the recurrence of the same sequence of state transitions under identical conditions. In this paper, reproducibility is treated as an **emergent property** that arises naturally from the triadic relation of **structure, interaction, and energy**.

1. Structure finite-izes the possibility space

→ It limits the range of transitions the system can take.

2. Interaction constrains transition sequences

→ It determines the direction in which the system evolves.

3. Energy drives those sequences

→ It enables the system to actually proceed along that direction.

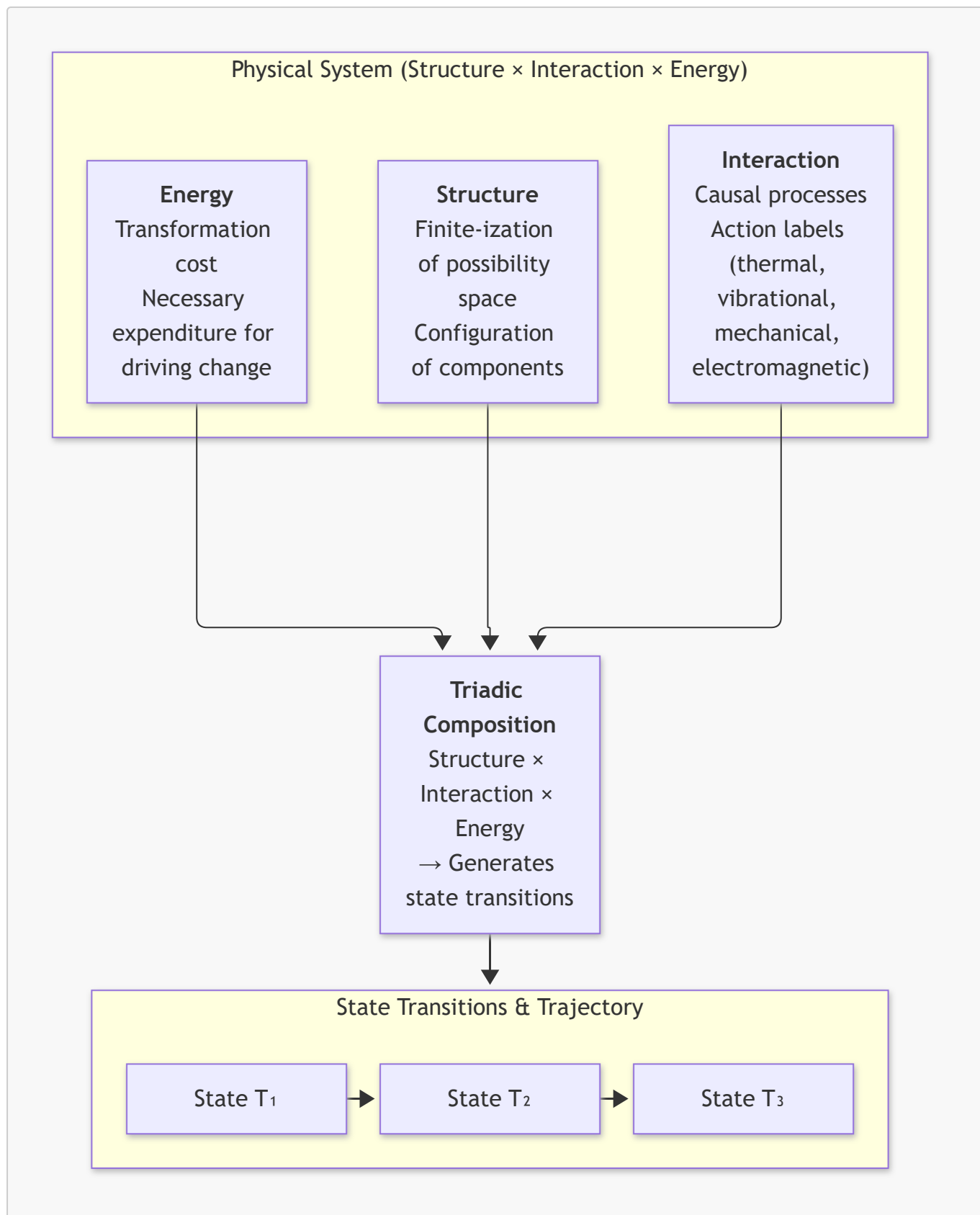
When these three elements are present, state transitions appear as **stable trajectories**. Structure defines the bounds, interaction provides directionality, and energy propels the system along that direction. Without this triadic relation, state transitions would diffuse in different directions each time, and reproducibility could not arise.

A crucial point here is that **the outcome of interaction × energy is a dynamic “state transition” (or an update to the next state), and this series of steps appears as observable points along a trajectory**. A trajectory manifests as a sequence of states, and the stability of this sequence forms the foundation of reproducibility in this paper.

The notion of reproducibility adopted here does not subordinate itself to the strict definitions used in mathematical physics. Instead, it is positioned as a **foundational perspective for unifying diverse phenomena—computation, cognition, language, and the distinction between known and unknown—through the stability of state transitions in the physical law domain**.

In the chapters that follow, this “reproducibility as stable trajectories” serves as the starting point for examining how states are mapped into other hierarchical layers, and how these mapping structures transform reproducibility.

Figure 1. Structure × Interaction × Energy → State Transitions (Diagram)



2.5 Fluctuation Is Not Noise but the Natural Behavior of the Possibility Space Allowed by Structure

The fluctuations treated in this paper are not errors or noise to be eliminated.

Fluctuation arises as a **natural variation in state transitions** originating from the breadth of the possibility space permitted by structure, and should be understood as an inherent property of structure itself.

- **Weakly constrained structure**
 - A wide range of permissible state transitions
 - Fluctuation appears large
- **Strongly constrained structure**
 - A narrow range of permissible state transitions
 - Fluctuation appears small

Both are “normal” behaviors of structure.

Fluctuation reflects the breadth of the possibility space defined by structure and does not undermine reproducibility.

Indeed, reproducibility is defined as **the stability of state transitions within the bounds permitted by structure**, and the width of those bounds manifests as observable fluctuation.

Although this paper does not analyze fluctuation in detail, it is important to state explicitly that fluctuation is not treated as error or noise. Fluctuation is not an exception to reproducibility; it is the **natural shadow of the possibility space inherent in structure**.

3. Partial Reproducibility in the Representational Domain: Structural Constraints as Limits of Mapping

Chapter 2 showed that reproducibility in the physical law domain **E** arises from the **triadic relation of structure × interaction × energy**.

However, the world a subject actually operates on is not this physical domain itself.

A subject cannot directly access the external world **E**.

Instead, the physical states of the external world are first mapped into internal physical states **P** through biological sensory processes:

$$F_{\text{sensor}} : \mathbf{E} \rightarrow \mathbf{P}$$

These internal physical states are then mapped into an internally structured representational state-space **I**:

$$\Phi : \mathbf{P} \rightarrow \mathbf{I}$$

Through this two-stage mapping, the world available to the subject becomes

$$\mathbf{I} = \Phi(F_{\text{sensor}}(\mathbf{E}))$$

a **representational domain closed under the image of biological sensors**.

In other words, the world the subject operates on is not the external world itself, but an **internal world bounded by what biological sensors can capture**.

This chapter examines how reproducibility—originally established in the physical domain—is transformed within the representational domain **I**, and why it appears as **partial reproducibility** under the structural constraints of mapping.

The mapping structure used here was formalized in *Categorical Internal World Model* (DOI: 10.5281/zenodo.22870778).

This paper assumes that structure and builds upon it.

3.1 Reproducibility in the Physical Domain Can Bring Only “States” into the Representational Domain

In the physical law domain, state transitions are determined by the triadic relation of **structure × interaction × energy**, and the **trajectory itself** is the core of reproducibility.

However, biological sensory mappings **cannot map trajectories**.

A biological sensor does not observe causal sequences themselves; it can only observe **a single state** that appears as the *result* of such a sequence.

This is because structure, interaction, and energy all possess properties that biological sensors cannot directly access:

- **Structure** → The configuration of components and the possibility space cannot be observed without disassembling the system.
- **Interaction** → Action labels such as thermal, vibrational, or electromagnetic are not directly observable; only the resulting state changes can be sensed.
- **Energy** → Contributes as transformation cost, but its “identity” cannot be directly observed.

Therefore, what is mapped into the representational domain are only:

- the **state** observed at some point along the trajectory, and
- the **fact** that this state did not break down within the physical domain’s structure.

In other words, what can be carried into the representational domain is:

only the resulting “state” that existed within the reproducibility of the physical domain,
not reproducibility itself (i.e., not the trajectory).

Concrete Example: Mapping the Phenomenon of a Falling Apple

Consider the physical-domain phenomenon “an apple falls from a tree.”

In the physical domain, this phenomenon is a **continuous trajectory** determined by gravity, air resistance, and structural constraints.

However, in the representational domain, this trajectory is not mapped.

Instead, only the following **discrete states** are taken in:

- **An apple on the branch (State 1)**
- **An apple on the ground (State 2)**

The causal structure—gravity, air resistance, dynamical constraints—is **not** mapped.

The representational domain can handle **state fragments**, not trajectories.

3.2 The Fact That a State Was Observed Guarantees a “Non-Breaking Transition”

When a state is observed in the physical domain, this observation implies that the state:

- existed without contradiction within the constraints of structure,

- did not violate the conditions of interaction and energy, and
- appeared somewhere within a valid sequence of state transitions.

A crucial point here is that **what enters the representational domain is only the resulting “state,” while all underlying causal structures—structure, interaction, energy, and trajectory—are not mapped.**

However, the mere fact that a state was observed guarantees at least one thing:

the transition leading to that state did not break down.

Thus, within the representational domain:

If two or more states are observed, it is guaranteed that a “non-breaking transition” existed between them.

Yet the following remain unmapped and therefore unknown:

- what specific interactions occurred,
- what energy conditions were involved,
- what trajectory was followed.

What enters the representational domain is only:

- the **fact of non-breakdown**, and
- the **resulting observed state**.

Concrete Example: Guarantee of a Non-Breaking Transition in Apple Falling

If “an apple on the branch (State 1)” and “an apple on the ground (State 2)” are observed, this fact implies that a **non-breaking transition** existed between them.

In the representational domain:

- how gravity acted,
- how much air resistance influenced the motion,
- what exact trajectory was followed,

none of these causal structures are mapped.

However:

It is guaranteed that a non-breaking transition existed from State 1 → State 2.

Thus, the fact that “the apple fell” is treated in the representational domain as an **existence guarantee (existence relation)**.

3.3 “Partial Reproducibility” in the Representational Domain

From the asymmetries described above, reproducibility in the representational domain can be defined as follows.

Definition (Partial Reproducibility)

Reproducibility in the representational domain is the condition in which:

- states observed in the physical domain
- can be placed side-by-side within the representational domain without contradiction, and
- it is guaranteed that a **non-breaking transition** existed between those states.

What is guaranteed here is **not** the causal structure that existed in the physical domain, but only:

- the **observed states** that resulted from that causal structure, and
- the **fact** that those states were physically non-breaking.

Therefore, this is not a full mapping of reproducibility in the physical domain. It is:

merely a reconstruction within the representational domain of fragments of states that existed under physical reproducibility.

Thus, reproducibility in the representational domain is **partial reproducibility**.

3.4 Mathematical Modeling of Partial Reproducibility

Within the representational domain $\mathbf{I}$, the structural constraint described in the previous sections—namely, that *only states are mapped*—can be formalized as follows.

(1) Mapping of States

Physical-domain states $e_1, e_2 \in \mathbf{E}$ are mapped into representational-domain states $i_1, i_2 \in \mathbf{I}$ through biological sensory mapping and internal-world mapping:

$$i_1 = \Phi(F_{\text{sensor}}(e_1)), \quad i_2 = \Phi(F_{\text{sensor}}(e_2)).$$

What is mapped here are **states only**.

Causal structures—structure, interaction, energy—are **not** mapped.

(2) “Non-Breaking Transition” as an Existence Guarantee

The fact that e_1 and e_2 were observed in the physical domain implies that:

- each state existed without contradiction within structural constraints,
- interaction and energy conditions did not break down, and
- both states appeared somewhere within a valid sequence of state transitions.

Therefore, in the representational domain, the following relation is guaranteed:

$$i_1 \rightsquigarrow i_2$$

This expresses the **existence of a non-breaking transition** between the two states.

What is guaranteed is **only the existence** of such a transition.
Its content—what interactions, what energy conditions, what trajectory—is completely lost.

(3) Partial Transition Sequences (Partial Trajectory)

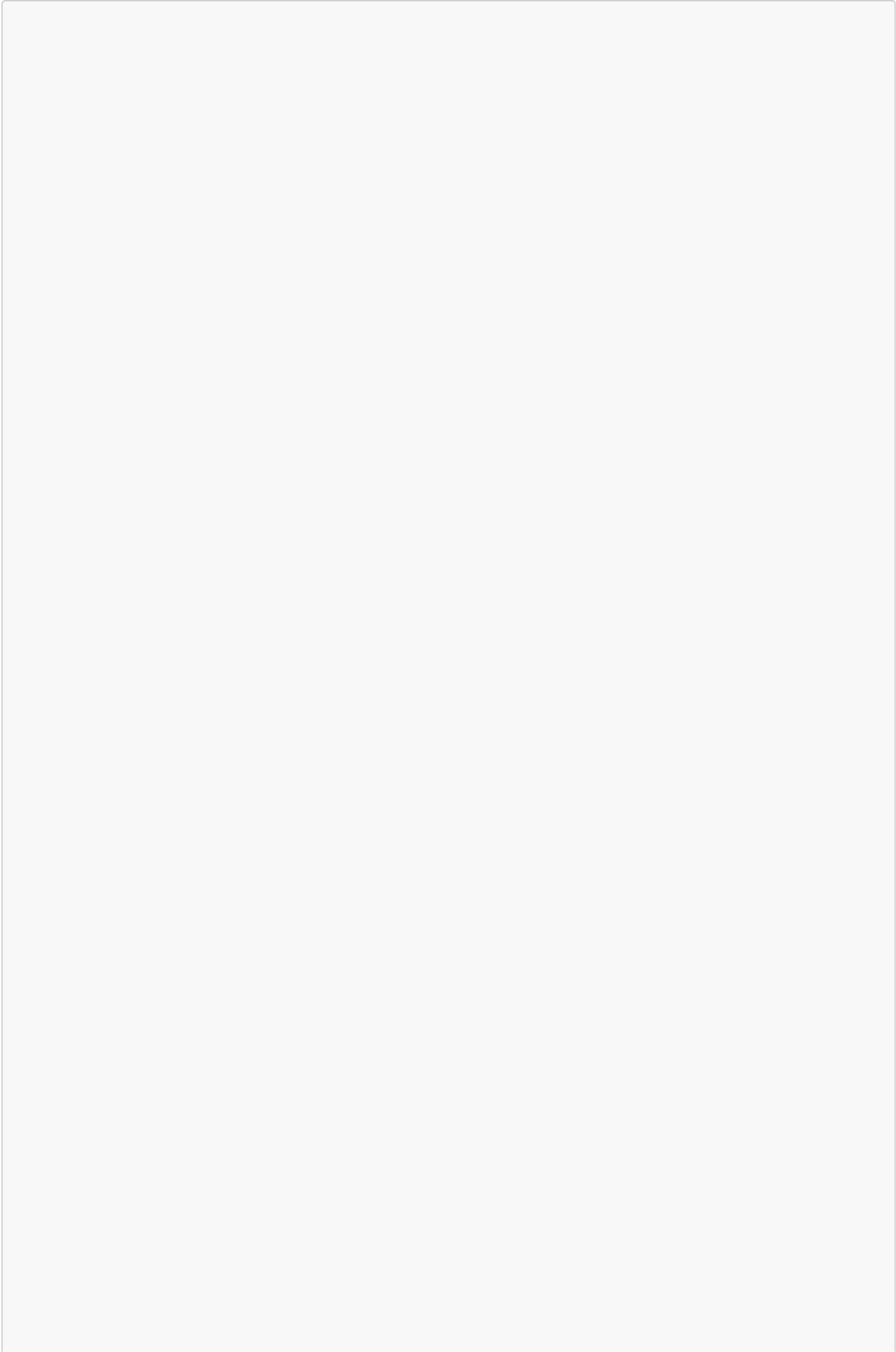
If multiple states are observed, one can construct:

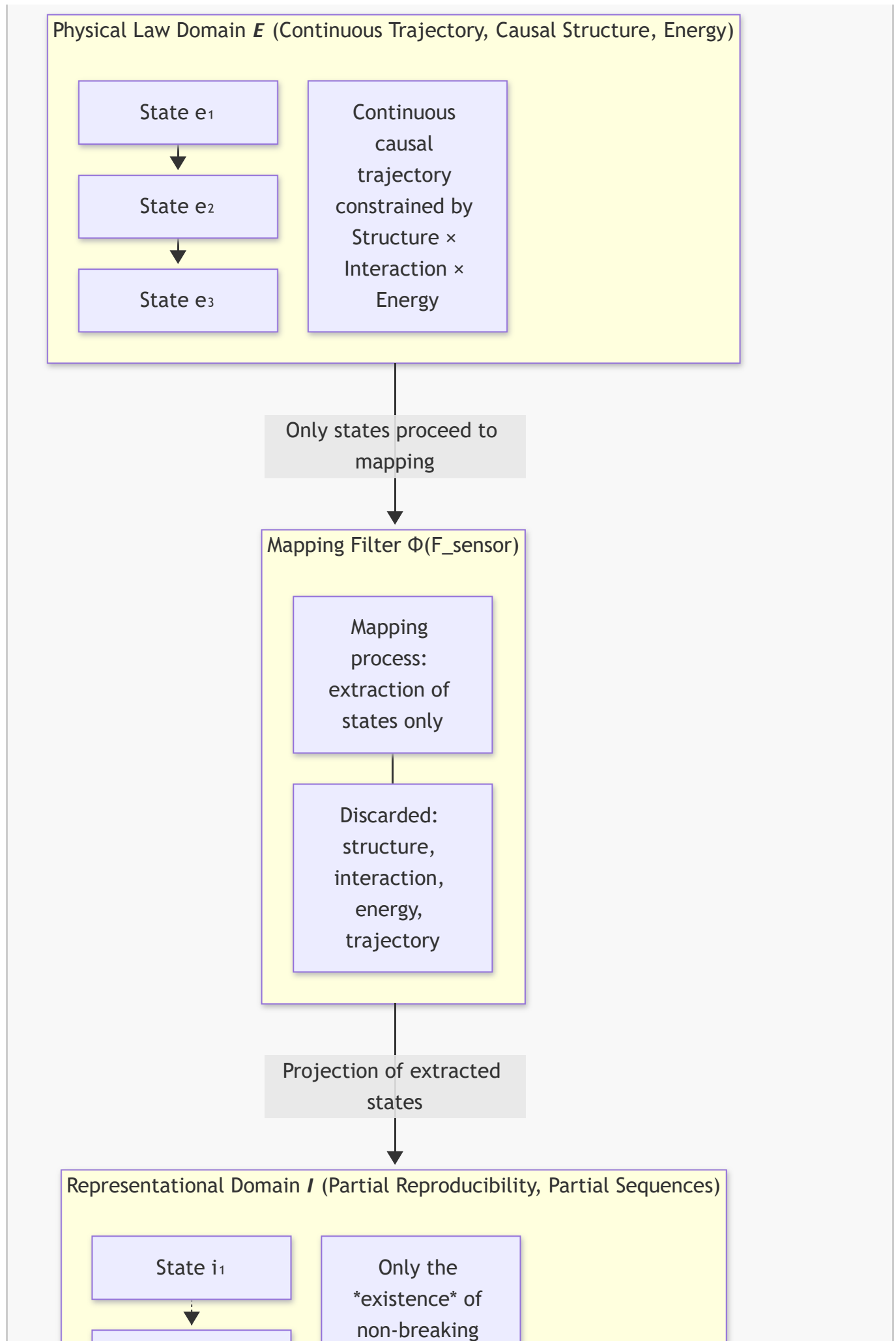
$$i_1 \rightsquigarrow i_2 \rightsquigarrow i_3$$

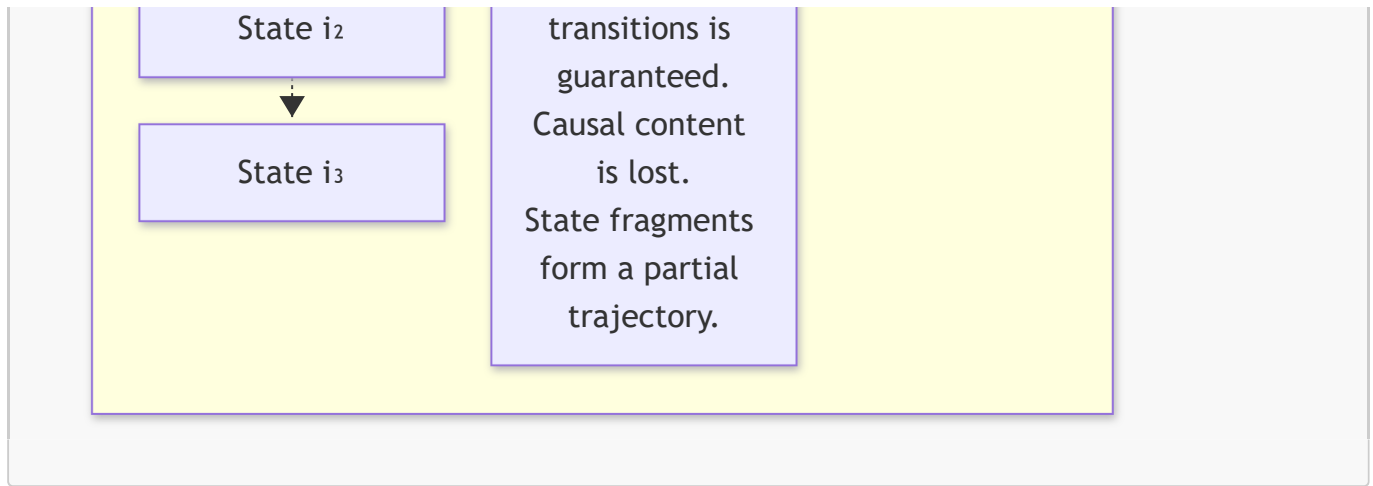
a “non-breaking sequence.”

However, this is **not** a full mapping of the physical-domain trajectory.
It is merely a **partial trajectory**, a chain of state fragments connected together.

Figure 2. Mapping from the Physical Law Domain to the Representational Domain and the Emergence of Partial Reproducibility





**Note:**

In this paper, we do not address methods for increasing the precision of partial reproducibility, nor methods for inferring or reconstructing the lost causal structure.

Instead, we focus exclusively on **the structural constraints of reproducibility within the representational domain**.

4. Structural Definition of Known (kichi) / Unknown (michi): Known as Reconstructibility in the Representational Domain

The concepts of known and unknown are widely used in cognitive science, information science, and the life sciences, yet their definitions remain consistently ambiguous. In particular, the question “*What does it mean for something to be known?*” appears frequently in discussions of symbol manipulation, learning, reasoning, and computation, but it has not been formalized in an implementable manner.

The Michi-Drive Hypothesis (DOI: 10.5281/zenodo.20990020) defined “known” as:

a state that possesses a reusable algorithm (state as entity).

However, this definition inevitably raises further questions:

- What is reutilization?
- What is an algorithm?
- Under what conditions can something be called reusable?
- Why do “reproducible” and “non-reproducible” entities exist?
- Is known/unknown a *property* of a state, or is it an *entity* in its own right?

To answer these questions, we must adopt the framework of **partial reproducibility in the representational domain** established in Chapter 3.

In the representational domain, trajectories of the physical domain are not mapped; only state fragments and the existence guarantee of non-breaking transitions are available.

Given this structural constraint, the need arises to redefine known/unknown as **entities** in terms of **reconstructibility (refrain)** within the representational domain.

This chapter reinterprets known (kichi) and unknown (michi) as **structural forms of reconstructibility in the representational domain**, grounded in partial reproducibility.

4.1 Known/Unknown Is Not a “Property” of a State but the State Itself (Entity) Permitted by Structure

A subject cannot directly operate on the external world.

It can only operate on states within the **representational domain $\mathbf{I}$** , obtained through the mapping:

$$\mathbf{E} \xrightarrow{F_{\text{sensor}}} \mathbf{P} \xrightarrow{\Phi} \mathbf{I}.$$

What is mapped into the representational domain are **states only**; the reproducibility of the physical domain (its trajectories) is not mapped.

Therefore, known/unknown is **not** a “property” of a state. It is:

the state itself (entity) as a point in the possibility space permitted by structure.

This follows from the reasons below.

(1) What is mapped into the representational domain is the state itself, not its properties

The mapping mechanism imports only the state (entity).

There is no mechanism that maps “known flags” or “unknown flags” as properties.

(2) The fact that a state was observed guarantees that it did not break down within the physical domain

Thus, what appears in the representational domain is a **non-breaking state as entity**.

(3) Whether a state can be reconstructed in the representational domain depends on whether the entity is compatible with the structure of that domain

Known/unknown is not a property but a question of **structural compatibility** (i.e., reconstructibility or refrain-possibility) within the representational domain.

Difference Between Attribute Interpretation and Entity Interpretation (Minimal Supplement)

Traditional “attribute-based” interpretations define known by attaching a label such as “known = True” to a state A .

In contrast, this paper treats known/unknown as **entities themselves**.

- **Attribute-based interpretation (traditional)**
 - Attach a “known flag” to state A .
 - Known is treated as a *property*.
- **Structural (entity-based) interpretation in this paper**
 - Determine whether state A can be reconstructed as $A \rightarrow A'$ without breakdown within the

representational domain.

→ Known is treated as **the entity itself that is reconstructible**.

Thus, known/unknown is defined not as a property of a state but as:

the structural compatibility inherent to the state itself (entity).

Concrete Example: Structural Interpretation of Known/Unknown in Apple Falling

As described in Chapter 3,

the physical-domain phenomenon “an apple falls from a tree” is a **continuous trajectory** determined by gravity, air resistance, and structural constraints.

However, in the representational domain, this trajectory is not mapped.

Only the following **discrete states (entities)** are imported:

- **An apple on the branch (State 1)**
- **An apple on the ground (State 2)**

In this case:

- State 1
- State 2
- The existence of a non-breaking transition from State 1 → State 2

are all reconstructible within the representational domain.

Thus, they are treated as **entities belonging to known (kichi)**.

This example concretely illustrates the central claim of this paper:

known/unknown is not a property of a state but an **entity determined by whether it can be reconstructed within the structure of the representational domain**.

4.2 Unknown (michi) Is Not Merely the Complement of Known (kichi)

When known/unknown is treated set-theoretically, it is often misunderstood as

“unknown is the complement of known.”

However, from the perspective of **operability (refrain-possibility) within the representational domain**, this interpretation is incorrect.

(1) Known = reconstructible states; Unknown = non-reconstructible states

Known/unknown is not a set-theoretic label.

It is determined by **whether a state can be reconstructed (refrained) within the representational domain**.

This is a classification based on operability,

and does not coincide with a set-theoretic complement.

(2) Many unobserved states exist, but they constitute only part of the unknown

The possibility space permitted by structure is finite,
and many states within it remain unobserved.

However, whether these states belong to the “complement of known” is a separate question:
their reconstructibility within the representational domain is not guaranteed.

(3) Unconfirmed transition paths also belong to the unknown, but not as a complement

The transition paths permitted by structure are finite,
and many of them remain unconfirmed.

These also belong to unknown (michi),
but again, they do not coincide with the complement of known.

(4) Unknown (michi) is a hierarchically broad entity

Thus, michi (unknown) consists of:

****states that cannot be reconstructed in the representational domain**

- states permitted by structure but not yet observed
- transition paths permitted by structure but not yet confirmed**

forming a **hierarchically broad entity**,
not merely the complement of known.

Concrete Example: Structural Interpretation of Unknown (michi) in Apple Falling

In §4.1,

“apple on the branch (State 1),”

“apple on the ground (State 2),”

and the non-breaking transition between them

were shown to be **reconstructible** within the representational domain—
thus belonging to **known (kichi)**.

However, the same phenomenon contains many entities that cannot be reconstructed:

- gravity as interaction
- air resistance as an action label
- the continuous falling trajectory
- continuous profiles of velocity and acceleration
- changes in airflow and pressure distribution

These are not directly mapped due to sensory bandwidth limitations,
and cannot be reconstructed within the representational domain.

Therefore, they belong to **unknown (michi)**.

Even if a subject can “mentally replay” the falling motion,
this replay is merely an **internally completed image permitted by the representational structure**,

not a reconstruction of the raw gravitational field or continuous causal dynamics.

Unknown (michi) also changes with the subject's stage of learning

Unknown is not a simple complement;
it changes hierarchically depending on the presence or absence of internal models.

(1) Unknown in the raw physical domain (permanently unknown)

Microscopic chaotic air resistance,
infinitely continuous trajectory data,
and the raw gravitational field
cannot be mapped due to sensory bandwidth limits.
No matter how much the subject learns,
these remain **permanently unknown (michi)** as physical entities.

(2) Unknown for a subject before learning (lack of causal models)

A subject who does not know gravity or air resistance
can only handle the non-breaking transition between State 1 and State 2.
The causal and sequential structure of falling cannot be reconstructed.
For such a subject, the falling process is **entirely unknown (michi)**.

(3) Boundary between known and unknown after learning or embodied experience

Through learning and embodied experience,
a subject constructs internal models such as a "gravity model"
or a "completion structure for continuous falling motion."

Although these models are not the raw gravitational field,
they can be reconstructed without breakdown within the representational domain.
Thus, the **completed falling motion becomes an entity of known (kichi)**.

Unknown (michi) is therefore not the remainder obtained by subtracting known.
It is a **hierarchical and dynamic entity** that spans:

- physically unknowable structures due to sensory limits,
 - unconstructed regions that may become known through learning,
 - and non-reconstructible states within the representational domain.
-

4.3 What Is Reuse? Reconstructibility in the Representational Domain

Known (kichi) is defined as a **state (entity) that is reusable**, but the notion of **reuse** here is not a mapping of reproducibility in the physical domain.

Rather, it is an operation that exists **within the internal structure of the representational domain**.

Reuse in the representational domain is:

the ability of a state (entity) to arise again without breakdown within the representational domain
(reconstructibility = refrain-possibility).

Note:

In this paper, *refrain* is not used in the general English sense of “refrain from.”

It is used as a technical term closer to the musical *refrain* (a recurring phrase) or the Deleuze–Guattari *ritournelle* (repetition / emergence).

It refers to the **re-arising (reactivation)** of a state within the representational domain.

This reconstructibility presupposes the **partial reproducibility** established in Chapter 3.

(1) Reuse is “reconstruction of a state,” not reproducibility in the physical domain

In the physical domain, state transitions are reproduced as continuous trajectories determined by structure × interaction × energy.

However, in the representational domain, trajectories are not mapped.

Only state fragments and the existence guarantee of non-breaking transitions are available.

Therefore, reuse in the representational domain is:

- not the reproduction of physical trajectories,
- but the re-arising (refrain) of a state
- on top of the internal structure of the representational domain.

(2) Reuse-possibility is determined by “internal structural compatibility”

A state is reusable if it is structurally compatible with the representational domain and can be reconstructed without breakdown.

If a state A :

- within the representational domain
- can be reconstructed as

$$A \rightarrow A' \rightarrow A''$$
without breakdown,

then the state is **reusable (kichi)**.

Conversely, states that cannot be reconstructed within the representational domain belong to **unknown (michi)**.

(3) Reuse is “reactivation of an internal model”

When a state is reconstructed,

the subject reactivates the **internal model (completion structure)** associated with that state.

The structural definition of these internal models is addressed in §4.4, but the essential point here is:

Reuse is the phenomenon in which an internally constructed model enables a state to arise again without breakdown.

Concrete Example: Reuse (kichi) in Apple Falling

As described in §4.1:

- apple on the branch (State 1)
- apple on the ground (State 2)
- the existence guarantee of a non-breaking transition between them

are all reconstructible within the representational domain, and thus belong to **known (kichi)**.

A subject can mentally replay “the apple falling” because the **completion model of falling motion** constructed within the representational domain can be reactivated.

This is a concrete example of:

reuse = reconstructibility (refrain).

4.4 What Is an Algorithm? Transition Sequences Constructed in the Representational Domain

An algorithm in the representational domain is:

a non-breaking sequence of state transitions that can be constructed within the representational domain
(a sequence built on existence relations).

This is not a full mapping of trajectories in the physical domain.

(1) Trajectories in the physical domain are not mapped

- Physical trajectories are determined by structure × interaction × energy.
- They cannot be observed by biological sensors.
- Therefore, they are not mapped into the representational domain.

What is carried into the representational domain are **only the state fragments observed as the result of the trajectory**.

(2) Algorithms in the representational domain are constructed by “placing state fragments side-by-side”

Within the representational domain, observed state fragments can be arranged as:

$$i_1 \rightsquigarrow i_2 \rightsquigarrow i_3$$

a **non-breaking transition sequence (partial trajectory)**.

This sequence is not a full mapping of the physical trajectory.

It is constructed on top of **partial reproducibility**.

(3) Internal models function as “representational-domain versions of structure × interaction × internal driving energy”

A crucial point is that the representational domain also contains **constraints corresponding to those of the physical domain**.

In the physical domain:

- **Structure**
- **Interaction**
- **Energy**

jointly constrain state transitions and form trajectories.

These are not mapped into the representational domain,
but the subject constructs **internal-model counterparts** of these constraints.

Triadic relation of the internal model (constraints of the representational domain)

- **Internal Structure**
 - configuration of the representational state-space
 - determines which states are compatible with which
- **Internal Interaction**
 - “possibility of sequencing” states
 - e.g., imagined falling motion, continuity of sound, grammatical transitions
- **Internal Driving Energy**
 - the “drive” required to reactivate states
 - attention, memory, embodied experience as reconstruction cost

When these three are present,
the representational domain can construct **non-breaking transition sequences (algorithms)**.

Thus, an algorithm is:

**a state sequence that can be driven without breakdown
by the internal version of structure × interaction × energy
within the representational domain.**

(4) An algorithm is the “operational range” of an internal model

Once an internal model is established, the subject can:

- reconstruct states,
- drive state sequences again,
- reuse those sequences in other contexts.

This **operational range**—the dynamically traversable, structurally valid region—is what constitutes an algorithm.

An algorithm is a **sequence that can be moved again (refrained) within the representational domain**.

Concrete Example: Algorithm of Falling Motion

In the apple example:

- State 1 (apple on the branch)
- State 2 (apple on the ground)
- the existence guarantee of the transition State 1 → State 2

are reconstructible within the representational domain.

In addition, through learning, the subject acquires:

- an internal interaction model of “falling,”
- internal structural compatibility for “gravity acting,”
- internal driving energy (attention, memory) to replay the falling image.

With these components,

the algorithm of falling motion (an internal transition sequence) becomes established within the representational domain.

This is not the physical trajectory itself,
but an **incomplete yet non-breaking sequence constructed internally**.

4.5 Conditions for Known/Unknown: kichi/michi as the Boundary of Reconstructibility

Known (kichi) and unknown (michi) are determined by **reconstructibility (refrain-possibility)** within the representational domain.

This section organizes where and how the boundary between known and unknown emerges from the structural viewpoint of the representational domain.

(1) Conditions for Reuse: Requirements for a State to Arise Again

For a state to be reconstructed within the representational domain, the following conditions must be satisfied.

① The representational domain must be closed

$$\mathbf{I} = \Phi(F_{\text{sensor}}(\mathbf{E}))$$

If the representational domain is not closed,
there exists no mapping chain through which a state can be reconstructed.

② The state must be reconstructible without breakdown

Within the representational domain, it must be possible to construct:

$$i_1 \rightsquigarrow i_2$$

a **non-breaking transition**.

③ The transition sequence must remain closed within the representational domain

Whether a sequence can be reconstructed as an algorithm depends on
whether the internal structure of the representational domain permits that sequence.

(2) Complete Definition of Known (kichi) / Unknown (michi)

Based on the above, known/unknown can be defined as follows.

Known (kichi)

A state (entity) is known (kichi) if it can be reconstructed without breakdown within the representational domain.

This means:

- the state is structurally compatible with the representational domain,
- its transition sequences can be reconstructed,
- it is reusable as an algorithm.

Unknown (michi)

A state (entity) is unknown (michi) if it breaks down during reconstruction within the representational domain, or if it has not yet been constructed.

However, michi is not a mere complement.

It is a **hierarchically broad entity** that includes:

- states that cannot be reconstructed within the representational domain,
- states permitted by structure but not yet observed,
- transition paths permitted by structure but not yet confirmed.

(3) Hierarchical Structure of Known/Unknown: Physical Domain → Representational Domain → Symbolic Domain

Known/unknown emerges across the following three-layer structure:

① Reproducibility in the physical domain (trajectory)

Structure × interaction × energy → genuine reproducibility → not mapped into the representational domain.

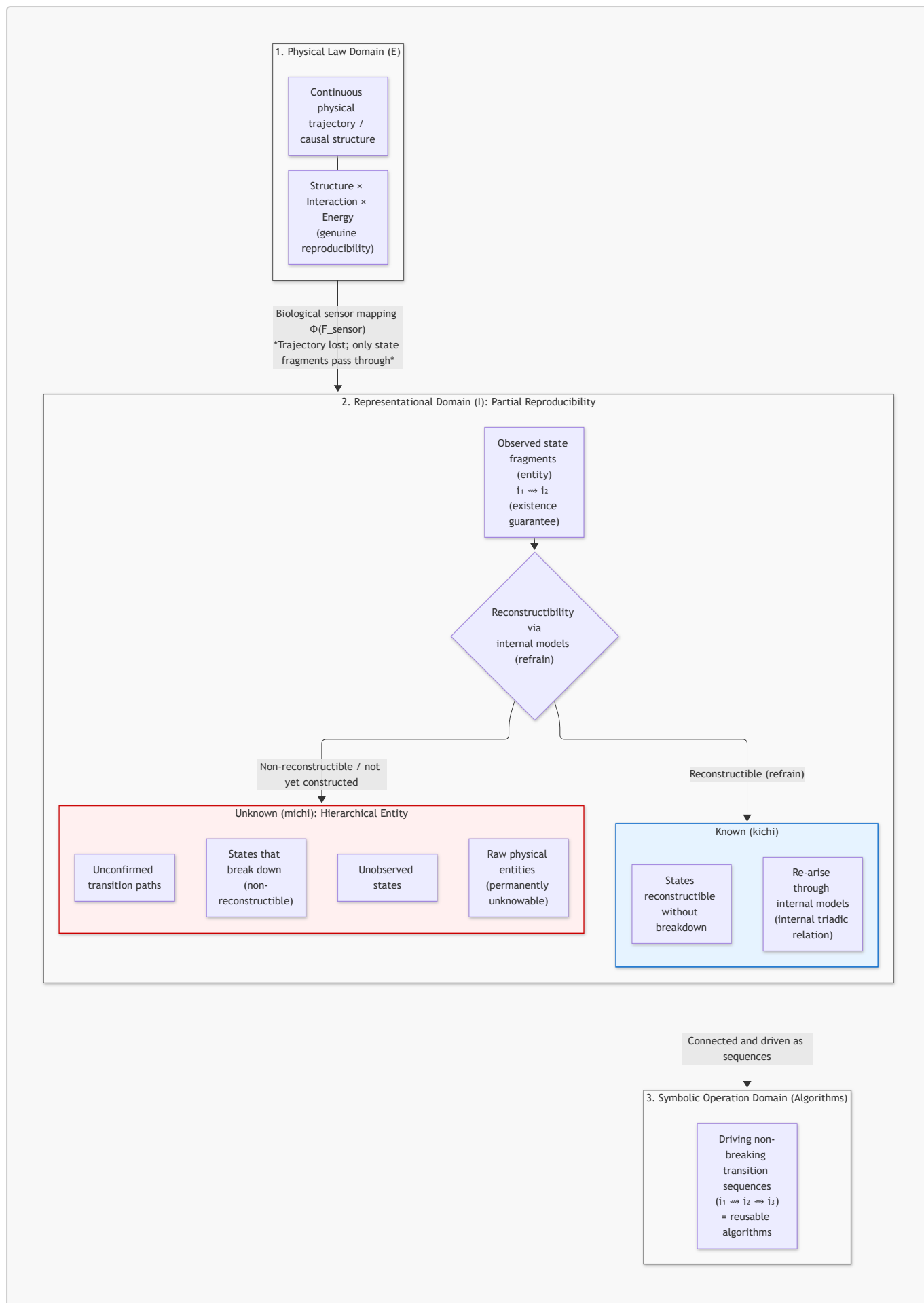
② Partial reproducibility in the representational domain (state fragments)

Only states are mapped → only non-breaking transitions are guaranteed → reconstructibility determines known/unknown.

③ Reusability in the symbolic-operation domain (algorithm)

Transition sequences constructed within the representational domain → reusable sequences are known (kichi)
→ non-reusable sequences are unknown (michi).

Figure 3. Structural Development of Known/Unknown Based on Reconstructibility (Refrain) in the Representational Domain



5. Connection to Symbolic and Linguistic Operations: Algorithms as Ordered Sequences of States

As established in Chapters 1–4, the world a subject operates on is not the physical law domain **E**, but the **representational domain I** obtained through biological sensory mapping and internal-world mapping.

Only *states* are mapped into the representational domain;

the reproducibility of the physical domain—its trajectories—is not mapped.

Consequently, reproducibility within the representational domain exists only as **partial reproducibility**.

On top of this partial reproducibility, the subject performs symbolic and linguistic operations.

Symbols and language do not operate on trajectories in the physical domain.

Rather, they operate on **state-transition sequences (algorithms)** constructed within the representational domain.

This chapter shows how symbolic and linguistic operations align with the structural constraints and reconstructibility of the representational domain.

5.1 Algorithms Are Ordered Sequences of States (Entities)

States in the representational domain include:

- symbols,
- sentences,
- contexts,
- linguistic units,
- internal cognitive states,

all of which are **entities—points in the state-space permitted by structural constraints**.

Symbols and sentences are not physical entities in the external world.

They arise as **discrete states** within the representational domain—states that are *detached from the continuity of physical spacetime* and treated as cognitive/representational nodes.

Because only states are mapped into the representational domain, the subject can operate only on **the placement and transition of states**.

For this reason, algorithms in the representational domain are defined as:

An algorithm is a transition sequence—an ordered arrangement of states (entities)—constructed within the representational domain.

The “ordering” here is not a full mapping of physical trajectories.

It is a **non-breaking sequence (partial trajectory)** constructed by placing observed state fragments side-by-side.

Symbolic and linguistic operations arise by manipulating these **non-breaking state sequences**.

Thus, linguistic meaning and symbolic manipulation are supported by:

the ordering of state transitions (algorithms) permitted within the representational domain.

5.2 Symbolic Operations Are Reconstructions of State-Transition Sequences

Symbolic operations are actions in which the subject manipulates states within the representational domain and **reconstructs new state-transition sequences (algorithms)**.

State transitions in the representational domain are not full mappings of physical trajectories. They are **non-breaking sequences (partial trajectories)** constructed by placing observed state fragments side-by-side.

Thus, symbolic operations are:

**operations that reconstruct state-transition sequences (algorithms)
on top of the partial reproducibility of the representational domain.**

Symbols are not physical entities.

They arise as **discrete states (entities)** within the representational domain.

Therefore, symbolic operations consist of placing symbolic states and constructing sequences whose ordering does not break down.

Concrete Example: Symbolic Operations Using “+” and “−”

As a minimal example, the operators “+” and “−” are treated within the representational domain as follows:

- “3”
- “+”
- “5”

When these three symbolic states are placed together, the subject constructs—based on internal models—the following **non-breaking transition sequence**:

$$3 \rightsquigarrow + \rightsquigarrow 5 \rightsquigarrow 8$$

The arrival at “8” is not a physical trajectory.

It is a **symbolic algorithm** produced by the **internal model (addition algorithm)** within the representational domain.

A subject can understand the symbolic string “3 + 5” because its ordering is compatible with the internal structure of the representational domain (including social conventions) and can be reconstructed without breakdown.

Constraints of Symbolic Systems: Social Convention

The essence of symbolic operations is the reconstruction of state-transition sequences.

However, **which symbols count as which states**, and **which orderings are considered non-breaking**, are determined not by physical laws but by **social conventions**.

Examples:

- "+" denotes addition
- "=" denotes equivalence
- "犬" denotes *dog*
- grammatical transitions are determined by linguistic convention

Thus, symbolic systems are established through **social rules that constrain the structure of the representational domain from the outside**.

Symbolic operations function by reconstructing state-transition sequences on top of these socially agreed-upon structural constraints.

Essence of Symbolic Operations

Symbolic operations consist of:

- selecting symbolic states,
- constructing an ordering,
- ensuring the sequence does not break down,
- activating meaning through internal models.

They are, fundamentally, **operations on state-transition sequences**.

This structure is continuous with the next section on **linguistic operations (sequences of sentences and contexts)**.

Symbolic operations constitute the minimal unit of linguistic operations.

5.3 Linguistic Operations Are the Reuse of Structured State Sequences

Linguistic operations arise as a higher-order structure built on top of symbolic operations.

Whereas symbolic operations reconstruct the ordering of discrete states (symbols), linguistic operations **structure state sequences contextually and hierarchically, and externalize them in a reusable form**.

Language combines:

- **states (entities),**
- **orderings of states (algorithms),**
- **structuring of sequences (sentences, contexts, discourse),**
- **reusability of sequences (kichi/michi)**

to form a coherent system.

(1) Linguistic operations manipulate "structured state sequences"

Symbolic operations handle local state transitions.

In contrast, linguistic operations manipulate the following hierarchical structure:

- symbol → word → phrase → sentence → context → discourse

This hierarchy is established by **structuring** state sequences within the representational domain.

(2) Linguistic operations reconstruct “contextual sequences”

Linguistic operations reconstruct not merely symbolic strings but **contextual sequences of states**.

Context is constrained by:

- preceding and following state sequences,
- the subject’s internal models,
- social conventions (pragmatics, grammar),
- the boundary between known and unknown.

Importantly, linguistic meaning is not “added” by context.

Rather:

the inherent polysemy (interpretability) of words and sentences is *restricted* by contextual interaction, narrowing the interpretive trajectory in the state-space into a single, low-uncertainty sequence that becomes reusable (kichi).

Context functions as a constraint that shapes a stable, reconstructible sequence.

(3) Linguistic operations are tightly coupled with “reusability (kichi)”

Because linguistic operations involve context, discourse, pragmatics, social conventions, and internal models, **the reusability (kichi/michi) of the entire sequence becomes a condition for meaning to arise**.

Linguistic operations reuse sequences belonging to known (kichi) and explore sequences belonging to unknown (michi).

(4) Linguistic operations are extensions of symbolic operations

Symbolic operations handle local state transitions.

Linguistic operations **structure, contextualize, and hierarchize** those sequences.

Thus, linguistic operations are:

- a higher-order structure built on symbolic operations,
 - a continuous extension of symbolic operations,
 - an elevation of symbolic reusability into a broader domain.
-

5.4 Reproducibility in Symbolic and Linguistic Operations: Social Convention as Structural Constraint

Symbolic and linguistic operations are state-transition sequences (algorithms) constructed on top of the **partial reproducibility** of the representational domain, as established in Chapters 2–4. However, their reproducibility is not guaranteed by the structure of the physical domain. It is **constrained by the social conventions of symbolic systems**.

This section clarifies the structural basis on which reproducibility in symbolic and linguistic operations is established.

(1) Reproducibility in Symbolic Operations: Strong Constraints from Social Convention

Symbolic operations place states (symbols) within the representational domain and reconstruct them as non-breaking transition sequences.

Which orderings count as “non-breaking” is determined not by physical laws but by **social convention**.

Examples:

- “+” denotes addition
- “=” denotes equivalence
- “ $3 + 5 \rightarrow 8$ ” is treated as a non-breaking sequence

All of these arise from **socially shared structural constraints**.

However, the “social convention” of mathematics is not arbitrary.

Mathematical symbolic systems are:

axiomatic consensuses—extremely rigid agreements designed to eliminate contradiction and preserve strict reproducibility.

Because of this, mathematical symbolic systems impose very strong constraints, the allowable range of state-transition sequences is narrow, and variability is minimal.

This strong constraint is what supports the high reproducibility of mathematics.

(2) Reproducibility in Linguistic Operations: Weak Constraints and a Broad Possibility Space

Natural language, in contrast, has weak constraints as a symbolic system.

- words are polysemous,
- sentences contain structural ambiguity,
- context does not “add” meaning but **restricts the range of interpretability**.

Thus, linguistic operations operate on a **much broader possibility space** than symbolic operations.

This breadth corresponds to the “fluctuation” described in §2.5:

- weak structural constraints → wide range of allowable transitions
- large apparent variability
- but this variability is not noise; it is natural structural fluctuation

Therefore, reproducibility in linguistic operations is:

**the ability to reconstruct state sequences without breakdown
within the range permitted by context, social convention, and internal models.**

Because natural language has weak constraints,
its variability is large and its reproducibility is lower than that of mathematics.

(3) Symbolic and Linguistic Operations Arise on the “Possibility Space Permitted by Structure”

Variability in symbolic and linguistic operations is not error or noise.

It is **natural fluctuation arising from the breadth of the possibility space permitted by the symbolic system.**

- Mathematics: strong constraints → narrow possibility space → small fluctuation
- Language: weak constraints → broad possibility space → large fluctuation

Both are structurally normal and do not undermine reproducibility.

Reproducibility means:

**state-transition sequences appear stably
within the range permitted by structural constraints (social convention).**

(4) Alignment with the Theory of Chapters 2–4

With this clarification, symbolic and linguistic operations can be positioned as follows:

- **Chapter 2:** Reproducibility in the physical domain = structure × interaction × energy
- **Chapter 3:** Reproducibility in the representational domain = non-breaking state transitions
- **Chapter 4:** Known/unknown = reconstructibility
- **Chapter 5:** Symbolic and linguistic operations =
state sequences constructed on partial reproducibility in the representational domain,
whose reproducibility is **constrained by social convention**

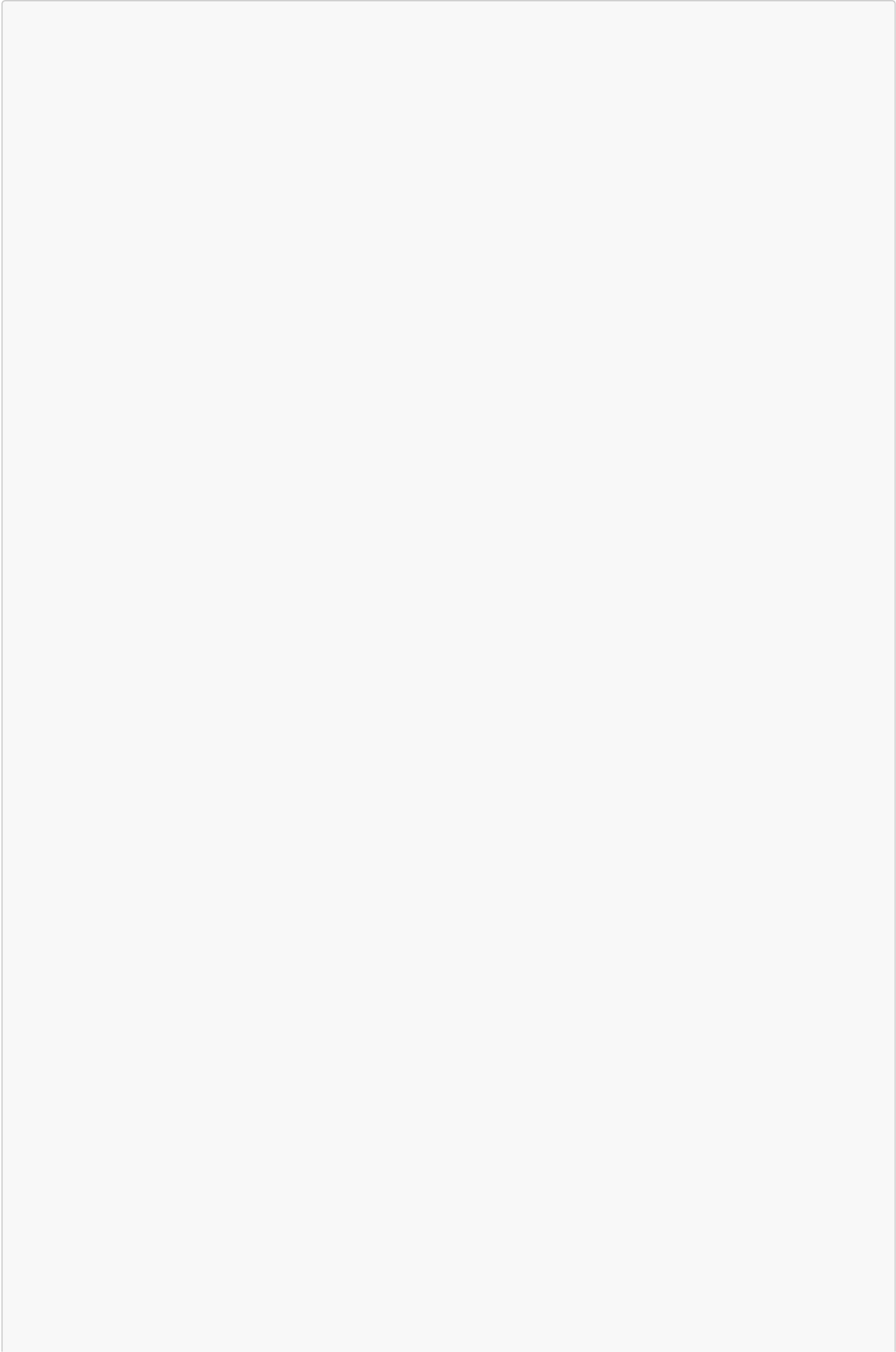
Thus, symbolic and linguistic operations align fully with the theoretical framework of Chapters 2–4.

Summary

**Reproducibility in symbolic and linguistic operations is the stable reconstruction of
state-transition sequences within the possibility space permitted by symbolic systems (social
convention).**

**Mathematics has strong constraints and high reproducibility;
natural language has weak constraints and large fluctuation.**

Figure 4. Comparison of Possibility Space and Reproducibility Based on the Strength of Social-Convention Constraints



Alignment with Theoretical Foundations (Chapters 2-4)

Chapter 2:
Reproducibility
in the physical
domain
(structure ×
interaction ×
energy)

Chapters 3-4:
Partial
reproducibility
in the
representational
domain
(non-breaking
state sequences
/ refrain)

Driven within the
representational domain

5.4 Reproducibility in Symbolic and Linguistic Operations (Strength of Social-Convention Constraints)

【Mathematics / Symbolic Systems】

Social
convention:
axiomatic,
rigorous
agreements
(strong
constraints)

Possibility
space:
extremely
narrow

Characteristics:
minimal
fluctuation
high
reproducibility
($3 + 5 \rightsquigarrow 8$)

【Natural Language Systems】

Social
convention:
pragmatics,
grammar,
context
(relatively
weak
constraints)

Possibility
space: broad
(polysemy,
ambiguity)

Characteristics:
meaning
constrained by
context
natural
variability
(fluctuation)
allowed

6. Conclusion: The Theoretical Ground Provided by the Perspective of Reproducibility

This paper aimed to redefine foundational concepts in cognition, information, and language—such as known/unknown, algorithms, and reusability—in an implementable manner by reframing them through a **two-layer structure of reproducibility**:

- (1) **reproducibility in the physical law domain**, and
- (2) **partial reproducibility in the representational domain**.

In Chapter 2, we showed that reproducibility in the physical law domain arises from the triadic relation of **structure × interaction × energy**.

This reproducibility concerns the *trajectory itself*, which the subject cannot directly operate on and which cannot be carried into the representational domain through biological sensory mapping.

In Chapter 3, we demonstrated that only *states* are mapped into the representational domain, and that the reproducibility of the physical domain (its trajectories) is not mapped.

Thus, reproducibility within the representational domain exists only as **partial reproducibility**, in which only *non-breaking transitions between states* are guaranteed.

In Chapter 4, we showed that on top of this partial reproducibility, **known (kichi) and unknown (michi) can be defined as entities**.

Known consists of states that are reconstructible within the representational domain, while unknown consists of states that are not reconstructible.

However, unknown is not merely the complement of known;

it is a **hierarchically broad entity** that includes states permitted by structure but not yet observed, as well as unconfirmed transition paths.

In Chapter 5, we demonstrated that this theoretical foundation allows symbolic and linguistic operations to be explained **as algorithms—ordered sequences of states**.

Symbolic and linguistic operations are “reconstructions of non-breaking state sequences” built on partial reproducibility in the representational domain.

Their reproducibility is determined not by physical laws but by **the social conventions of symbolic systems**.

Mathematical symbolic systems are constrained by **strong axiomatic consensus**, which eliminates contradiction and yields high reproducibility.

Natural language, by contrast, has weaker constraints, a broader possibility space for allowable state transitions, greater fluctuation, and a wider interpretive range of reproducibility.

6.1 The Theoretical Ground Provided by This Paper

This paper has presented the following unified structure:

- **Reproducibility in the physical domain**
→ trajectories determined by structure × interaction × energy

- **Partial reproducibility in the representational domain**
→ only states are mapped; only non-breaking transitions are guaranteed
- **Structural definition of known/unknown**
→ known (kichi) as reconstructibility
→ unknown (michi) as non-reconstructibility
- **Algorithms as ordered sequences of states**
→ symbolic and linguistic operations understood as reconstructions of state sequences
- **Constraints of symbolic systems (social convention)**
→ mathematics exhibits strong constraints and high reproducibility
→ natural language exhibits weak constraints, broad fluctuation, and wide interpretive variability

This theoretical ground provides a foundation for treating diverse phenomena in cognition, language, computation, learning, and reasoning as **questions of structural compatibility within the representational domain**.

6.2 The Physical Basis of Symbolic Operations: ATP and Structural Constraints

It is important to emphasize that the symbolic and linguistic operations discussed in this paper do **not** "float" as purely abstract representational manipulations.

Whenever a state changes within the representational domain, a corresponding change **must** occur in the physical law domain behind it.

When a subject performs symbolic operations, there are always:

- **energy expenditures** (ATP consumption, electrical activity), and
- **structural constraints arising from interactions among components** (neural circuits, electronic circuits, etc.)

present in the background.

Just as a calculator implements computation through electricity (energy) and circuitry (interaction constraints), a subject implements symbolic operations through **ATP (energy)** and the **structural constraints of internal models** (algorithms in the representational domain).

This perspective is often overlooked when symbolic or linguistic operations are treated solely as abstract processes.

Interpretation is left to the reader, but recognizing that operations in the representational domain are **always grounded in the constraints and energy consumption of the physical law domain** offers a new angle for understanding symbolic manipulation and computation.

6.3 Future Directions: Computational Gains Enabled by Partial Reproducibility

The perspective of **physical energy consumption and structural constraints** connects the theoretical foundation developed in this paper to a further question:

namely, **how a subject with only finite energy (ATP) can generate new state-transition sequences (algorithms) within its internal world and externalize them**.

The crucial point is that:

**partial reproducibility in the representational domain is not a defect;
it is a structural advantage that reduces computational cost.**

Unlike the physical domain or mathematical symbolic systems, a subject does not need to “compute” every strong constraint (axioms, strict rules) sequentially. By relying on **non-breaking state-transition sequences** that arise within the representational domain, the subject can bypass the full physical or logical exploration that would otherwise be required, and can generate new sequences (hypotheses, inferences, operations) at a fraction of the cost.

In other words:

- the structurally permitted possibility space of unknown (michi),
- partial reproducibility in the representational domain,
- reconstructibility of state sequences,
- stable sequences belonging to kichi,
- and the generation of new sequences (dynamic algorithm formation)

can all be treated as a unified process through which the subject forms hypotheses, externalizes them, and explores new structural configurations.

This paper does not delve into the full details, but the theoretical ground presented here naturally extends toward a formal theory of:

**how partial reproducibility reduces computational cost
and how subjects generate new algorithms.**

The next paper will attempt to describe this “bypassing of computational cost” through the hierarchical structure of reproducibility.

6.4 Conclusion

By reframing known/unknown through the perspective of reproducibility, this paper has established a foundation for treating diverse phenomena—symbolic operations, linguistic operations, internal models, computation, and reasoning—as **questions of structural compatibility within the representational domain.**

The framework presented here overcomes the vague definitions of known/unknown and provides a new theoretical ground for handling reusability, algorithms, and state transitions in an implementable manner.

Equally important is the recognition that:

**partial reproducibility in the representational domain is not a defect;
it is a structural advantage that enables the subject to bypass computational cost
and generate new state-transition sequences.**

There is no need to sequentially compute the strict trajectories of the physical domain or the strong constraints of mathematical symbolic systems (axiomatic structures).

By relying on **non-breaking state-transition sequences** that arise within the representational domain, the subject can conserve computational resources, explore the possibility space of the unknown, and generate new structures.

Formalizing this **computational gain enabled by partial reproducibility** is the natural next step built upon the theoretical ground established in this paper.

How the subject explores the unknown, generates new algorithms, and externalizes hypotheses—this further inquiry will be the focus of the next paper.
